



Bitcoining from space

Blockstream, a Bitcoin company announced the launch of a new satellite network that beams the digital currency for free. <http://dgit.in/SpaceCoin1>



Spider silk to kevlar

Researchers in the country have figured out a way to weld spider webs to kevlar using laser. <http://dgit.in/LaserWeb>

whole genome sequencing to find off-target effects may be missing potentially important mutations," Dr. Tsang says. "Even a single nucleotide change can have a huge impact." They hope that this study as well as their findings allow people to work on figuring out how to make CRISPR-Cas9 more specific and safer while editing genetic code. This is but a minor obstacle for a bright future.

Y'ALL GOT ANYMORE OF THOSE DESIGNER BABIES?

Ah, designer babies. The removal of the pseudo RNG element out of childbirth and make it an RPG character creation simulation instead – "Ah yes, I would like my daughter to have +20 Speed and +42 Beauty please. And go easy on the Derp Factor". A lot of new parents and parents-to-be would probably love to have this, and have this now and obviously, like literally everything else, the idea is also mired in controversy.

Childbirth in and of itself is a very polarising topic. When you add choosing characteristics into the mix, the polarisation goes up a notch. And there are valid arguments on both sides. The pro Designer Baby side argues that this could be the eventual, most beneficial evolution of mankind, making all babies supah smart, supah strong and supah beautiful. The collective IQ of the entire world goes up and the age old theory of the hot-crazy curve goes out the window.

Similarly, the Anti Designer Baby side say that it is going to create a horribly imbalanced world, where rich people's children have a distinct physical advantage due to them being able to afford gene therapy, and thereby create one of the most visual and easy to see discriminations the world has ever seen after the Apartheid. Or on the flip side, if literally everyone got access to them in some years, it could reduce diversity and make everyone copies of everyone else. A standardised (dare we say communistic?) world, where everyone is equally beautiful, equally smart and no one has any of their special quirks or talents. We aren't going to take any

sides here, but what we ARE going to say is, this argument is pointless. At least for now. CRISPR-Cas9 (Yup, you're going to be hearing about this a lot) while being quite effective, is still in it's infancy. We are able to isolate and work on specific SINGLE genes at max, while each of the human traits such as intelligence depend on multitudes upon multitudes of genes that we don't even know about. Even something very linear like height (whose cause and genes we are still learning about) could be caused by the combination of over 93,000 genetic mutations! For the record, we have identified only 700 of them. Even more damning to the cause is that CRISPR can't introduce new genetic code to the code it 'snips away'. The DNA automatically copies code from another part of the cell, so while this is extremely useful in cutting out diseased or troublesome parts, it makes introducing even a small mutation or new piece of code an extremely tall order. This is also a huge drawback to CRISPR because the single cell limitation means that we can only 'cure' diseases that occur due to a single specific gene mutation – thankfully there are enough of those, like for example Huntington's Disease and sickle cell anemia. CRISPR isn't going to advance to the level of combating the more complex diseases for a while yet, let alone mess around with human traits. Designer babies, for the better or worse, are still far off away. (<http://dgit.in/PradaBaby>).

FORGET IMAGES, PRINT VIRUSES

Synthetic Genomics, a synthetic biology company based out of California has unveiled a machine that is capable of printing out viruses. No, not the computer kind. They call it the Digital to Biological Convertor (very imaginative name, we know). It uses a module that is already widely available called the BioXP 3200, which researchers use to digitally synthesise DNA. The Digital to Biological Convertor can use this digital DNA to assemble parts of the virus together and essentially 'print' it out. This has numerous uses – people can send samples of viruses digitally

from places where there aren't facilities available, we could send the machine to Mars for example and get it to print organisms there to test out hypotheses, the possibilities are endless. Digitally synthesised DNA can be a huge boost to remote medicine and if at a place equipped with the proper CRISPR-Cas9 tech, doctors from the other side of the globe will be able to 'operate' on you on the genetic code level in the future.

SPACE BEAR PROTEIN, GOOD!

Scientists working over at the University of Tokyo have discovered that among all its varied remarkable abilities, Tardigrades (henceforth to be referred to as Space Bears... we mean, have you looked at them!?) have a special protein in their cells that protect them against X-Rays. But what's cooler is that if you transfer this protein, human cells ALSO get the radiation shield. How cool is that?

This protein, called DSUP prevents the Tardigrade's cell from disintegrating into mush when irradiated. Since such proteins are much easier to study



These little Tardigrades are cute, but tough

when inserted into a mammalian cell, researchers tried a human cell culture and inserting DSUP into it. It ended up reducing about 40% of the damage against head on X-Ray irradiation. This opens up new ways to enhance radiation therapy for cancer patients by increasing the cell's resistance level to the radiation. And the fact that this is just one protein of many in the Tardigrade genome, makes it all the more interesting. What more secrets does it hold? 